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# =====
# IMAGE REPRESENTATION STUDY USING OPENCV
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# Install libraries (run once in Colab)
!pip install opencv-python matplotlib numpy

# Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# -----
# STEP 1: UPLOAD IMAGE
# -----

print("Please upload an image file (jpg, png, jpeg)")
uploaded = files.upload()

# Get uploaded filename
image_path = next(iter(uploaded))

# -----
# STEP 2: READ IMAGE
# -----

image = cv2.imread(image_path)

if image is None:
    print("Error: Could not load image.")
    exit()

# Convert BGR to RGB for displaying
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# -----
# STEP 3: CONVERT TO GRAYSCALE
# -----

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# -----
# STEP 4: CONVERT TO BINARY
# -----

threshold_value = 127

_, binary = cv2.threshold(
    gray,
    threshold_value,
    255,
    cv2.THRESH_BINARY
)

# -----
# STEP 5: SAVE OUTPUTS
# -----

cv2.imwrite("grayscale_image.jpg", gray)
cv2.imwrite("binary_image.jpg", binary)

print("Grayscale image saved as grayscale_image.jpg")
print("Binary image saved as binary_image.jpg")

# -----
# STEP 6: DISPLAY RESULTS
# -----

plt.figure(figsize=(15,5))

# Original Image
plt.subplot(1,3,1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

# Grayscale Image
plt.subplot(1,3,2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")

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plt.figure(figsize=(10,10))
plt.axis("off")

# Binary Image
plt.subplot(1,3,3)
plt.imshow(binary, cmap="gray")
plt.title("Binary Image")
plt.axis("off")

plt.tight_layout()
plt.show()

# -----
# STEP 7: DOWNLOAD OUTPUT FILES
# -----

print("\nDo you want to download the generated images?")

files.download("grayscale_image.jpg")
files.download("binary_image.jpg")

```

Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
Please upload an image file (jpg, png, jpeg)

input co3 at 1 q2.jpg

input co3 at 1 q2.jpg(image/jpeg) - 169090 bytes, last modified: 10/8/2026 - 100% done

Saving input co3 at 1 q2.jpg to input co3 at 1 q2.jpg

Grayscale image saved as grayscale_image.jpg

Binary image saved as binary_image.jpg

Original Image



Grayscale Image



Binary Image



Do you want to download the generated images?

